

Lista03_Shiguihara

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Configurando o Rstudio

```
library(DBI)
library(RSQLite)

con <- dbConnect(SQLite(), "me315.sqlite")
```

Hands-On

1. Exiba os valores de STATE e FEE da tabela PRESERVE mas sem duplicados.

```
SELECT DISTINCT STATE, FEE FROM PRESERVE
```

Table 1: 4 records

STATE	FEE
AZ	3
MT	0
AZ	0
MA	0

2. Exiba todos os valores de DNO na tabela EMPLOYEE. Não exiba valores duplicados.

```
SELECT DISTINCT DNO FROM EMPLOYEE
```

Table 2: 3 records

DNO
20
10
40

3. Execute as QUERYS a seguir. Examine as tabelas resultantes e faça as observações relevantes.

- SELECT DNO, SALARY FROM EMPLOYEE
- SELECT DISTINCT DNO, SALARY FROM EMPLOYEE
- SELECT DISTINCT DNO, SALARY FROM EMPLOYEE ORDER BY DNO, SALARY Resposta: Como estamos selecionando duas colunas ao mesmo tempo, quando usamos o DISTINCT a função apenas tira as linhas repetidas, mas como não há nenhuma linha em que existe DNO e SALARY iguais ao mesmo tempo, nenhuma linha é retirada.

```
SELECT DNO, SALARY FROM EMPLOYEE
```

Table 3: 6 records

DNO	SALARY
20	2000
10	2000
20	3000
40	500
10	400
20	9000

```
SELECT DISTINCT DNO, SALARY FROM EMPLOYEE
```

Table 4: 6 records

DNO	SALARY
20	2000
10	2000
20	3000
40	500
10	400

DNO	SALARY
20	9000

```
SELECT DISTINCT DNO, SALARY FROM EMPLOYEE
ORDER BY DNO, SALARY
```

Table 5: 6 records

DNO	SALARY
10	400
10	2000
20	2000
20	3000
20	9000
40	500

4. Exiba todas as informações sobre qualquer reserva natural em Montana que seja menor que 1.000 acres.

```
SELECT * FROM PRESERVE
WHERE STATE = 'MT' AND ACRES < 1000
```

Table 6: 2 records

PNO	PNAME	STATE	ACRES	FEE
3	DANCING PRAIRIE	MT	680	0
40	SOUTH FORK MADISON	MT	121	0

5. Exiba todas as informações sobre qualquer reserva natural cujo valor de ACRES esteja entre 1200 e 15000, inclusive.

```
SELECT *FROM PRESERVE
WHERE ACRES >= 1200 AND ACRES <=15000
```

Table 7: 2 records

PNO	PNAME	STATE	ACRES	FEE
2	PINE BUTTE SWAMP	MT	15000	0
6	PAPAGONIA-SONOITA CREEK	AZ	1200	3

6. Exiba todas as informações sobre qualquer reserva natural localizada em Montana que não possua taxa de entrada e seja maior que 10.000 acres.

```
SELECT * FROM PRESERVE
WHERE FEE = 0 AND ACRES >10000
```

Table 8: 2 records

PNO	PNAME	STATE	ACRES	FEE
7	MULESHOE RANCH	AZ	49120	0
2	PINE BUTTE SWAMP	MT	15000	0

7. Exiba todas as informações sobre as reservas naturais localizadas em Montana ou Massachusetts.

```
SELECT * FROM PRESERVE
WHERE STATE = 'MT' OR STATE = 'MA'
```

Table 9: Displaying records 1 - 10

PNO	PNAME	STATE	ACRES	FEE
3	DANCING PRAIRIE	MT	680	0
40	SOUTH FORK MADISON	MT	121	0
14	MCELWAIN-OLSEN	MA	66	0
13	TATKON	MA	40	0
9	DAVID H. SMITH	MA	830	0
11	MIACOMET MOORS	MA	4	0
12	MOUNT PLANTAIN	MA	730	0
1	COMERTOWN PRAIRIE	MT	1130	0
2	PINE BUTTE SWAMP	MT	15000	0
10	HOFT FARM	MA	90	0

8. Exiba todas as informações sobre qualquer reserva natural localizada em Montana ou qualquer reserva que seja menor que 1.000 acres.

```
SELECT * FROM PRESERVE
WHERE STATE = 'MT' OR ACRES < 1000
ORDER BY STATE
```

Table 10: Displaying records 1 - 10

PNO	PNAME	STATE	ACRES	FEE
5	HASSAYAMPA RIVER	AZ	660	3
80	RAMSEY CANYON	AZ	380	3
14	MCELWAIN-OLSEN	MA	66	0
13	TATKON	MA	40	0
9	DAVID H. SMITH	MA	830	0
11	MIACOMET MOORS	MA	4	0
12	MOUNT PLANTAIN	MA	730	0
10	HOFT FARM	MA	90	0
3	DANCING PRAIRIE	MT	680	0
40	SOUTH FORK MADISON	MT	121	0

9. Exiba o número e o nome de qualquer reserva natural que possua taxa de entrada que não seja igual a zero.

```
SELECT PNO, PNAME FROM PRESERVE
WHERE NOT FEE = 0
```

Table 11: 3 records

PNO	PNAME
5	HASSAYAMPA RIVER
80	RAMSEY CANYON
6	PAPAGONIA-SONOITA CREEK

10. Exiba o número e o nome das reservas naturais que não possuem taxa de entrada de US\$ 3,00 e não possuem taxa de US\$ 10,00.

```
SELECT PNO, PNAME, FEE FROM PRESERVE
WHERE NOT (FEE = 3 OR FEE = 10)
```

Table 12: Displaying records 1 - 10

PNO	PNAME	FEE
3	DANCING PRAIRIE	0
7	MULESHOE RANCH	0
40	SOUTH FORK MADISON	0
14	MCELWAIN-OLSEN	0
13	TATKON	0
9	DAVID H. SMITH	0
11	MIACOMET MOORS	0
12	MOUNT PLANTAIN	0
1	COMERTOWN PRAIRIE	0
2	PINE BUTTE SWAMP	0

11. Exiba todas as informações sobre qualquer reserva natural localizada no Arizona que não possua taxa de entrada, ou qualquer reserva que seja menor que 100 **acres** (independentemente de seus valores de **STATE** e **FEE**).

```
SELECT * FROM PRESERVE
WHERE (STATE = 'AZ' AND FEE = 0) OR ACRES < 100
```

Table 13: 5 records

PNO	PNAME	STATE	ACRES	FEE
7	MULESHOE RANCH	AZ	49120	0
14	MCELWAIN-OLSEN	MA	66	0
13	TATKON	MA	40	0
11	MIACOMET MOORS	MA	4	0
10	HOFT FARM	MA	90	0

12. Exiba todas as informações sobre qualquer reserva natural que seja menor que 1.000 **acres** e tenha taxa de entrada de zero dólares ou esteja localizada no **Arizona**.

```
SELECT * FROM PRESERVE
WHERE ACRES <1000 AND FEE = 0 OR STATE = 'AZ'
```

Table 14: Displaying records 1 - 10

PNO	PNAME	STATE	ACRES	FEE
5	HASSAYAMPA RIVER	AZ	660	3
3	DANCING PRAIRIE	MT	680	0
7	MULESHOE RANCH	AZ	49120	0
40	SOUTH FORK MADISON	MT	121	0
14	MCELWAIN-OLSEN	MA	66	0
13	TATKON	MA	40	0
9	DAVID H. SMITH	MA	830	0
11	MIACOMET MOORS	MA	4	0
12	MOUNT PLANTAIN	MA	730	0
80	RAMSEY CANYON	AZ	380	3

13. Selecione todas as informações sobre qualquer reserva natural com taxa de entrada que não seja maior que zero, ou qualquer outra reserva, independentemente de sua taxa, que esteja localizada em **Montana** e seja maior que **1.000 acres**.

```
SELECT * FROM PRESERVE
WHERE FEE = 0 OR (STATE = 'MT' AND ACRES > 1000)
```

Table 15: Displaying records 1 - 10

PNO	PNAME	STATE	ACRES	FEE
3	DANCING PRAIRIE	MT	680	0
7	MULESHOE RANCH	AZ	49120	0
40	SOUTH FORK MADISON	MT	121	0
14	MCELWAIN-OLSEN	MA	66	0
13	TATKON	MA	40	0
9	DAVID H. SMITH	MA	830	0
11	MIACOMET MOORS	MA	4	0
12	MOUNT PLANTAIN	MA	730	0
1	COMERTOWN PRAIRIE	MT	1130	0
2	PINE BUTTE SWAMP	MT	15000	0

14. Exiba todas as informações sobre todas as reservas naturais, exceto aquelas reservas de Montana sem taxa de entrada.

```
SELECT * FROM PRESERVE
WHERE NOT STATE = 'MT' AND FEE = 0
```

Table 16: 7 records

PNO	PNAME	STATE	ACRES	FEE
7	MULESHOE RANCH	AZ	49120	0
14	MCELWAIN-OLSEN	MA	66	0
13	TATKON	MA	40	0
9	DAVID H. SMITH	MA	830	0
11	MIACOMET MOORS	MA	4	0
12	MOUNT PLANTAIN	MA	730	0
10	HOFT FARM	MA	90	0

15. As seguintes cláusulas WHERE são logicamente equivalentes?

Resposta: Sim, as cláusulas são logicamente equivalentes.

```
SELECT * FROM PRESERVE
WHERE STATE = 'MA' AND (ACRES > 1000 OR FEE = 0.0)
```

Table 17: 6 records

PNO	PNAME	STATE	ACRES	FEE
14	MCELWAIN-OLSEN	MA	66	0
13	TATKON	MA	40	0
9	DAVID H. SMITH	MA	830	0
11	MIACOMET MOORS	MA	4	0
12	MOUNT PLANTAIN	MA	730	0
10	HOFT FARM	MA	90	0

```
SELECT * FROM PRESERVE
WHERE (STATE = 'MA' AND ACRES > 1000) OR (STATE = 'MA' AND FEE = 0)
```

Table 18: 6 records

PNO	PNAME	STATE	ACRES	FEE
14	MCELWAIN-OLSEN	MA	66	0
13	TATKON	MA	40	0
9	DAVID H. SMITH	MA	830	0
11	MIACOMET MOORS	MA	4	0
12	MOUNT PLANTAIN	MA	730	0
10	HOFT FARM	MA	90	0

16. As seguintes cláusulas WHERE são logicamente equivalentes?

Resposta: Não, as cláusulas não são logicamente equivalentes.

```
SELECT * FROM PRESERVE
WHERE STATE = 'MA' OR (ACRES > 1000 AND FEE = 0.0)
```

Table 19: 9 records

PNO	PNAME	STATE	ACRES	FEE
7	MULESHOE RANCH	AZ	49120	0
14	MCELWAIN-OLSEN	MA	66	0
13	TATKON	MA	40	0
9	DAVID H. SMITH	MA	830	0
11	MIACOMET MOORS	MA	4	0
12	MOUNT PLANTAIN	MA	730	0
1	COMERTOWN PRAIRIE	MT	1130	0
2	PINE BUTTE SWAMP	MT	15000	0
10	HOFT FARM	MA	90	0

```
SELECT * FROM PRESERVE
WHERE (STATE = 'MA' OR ACRES > 1000) AND (STATE = 'MA' OR FEE = 0.0)
```

Table 20: 9 records

PNO	PNAME	STATE	ACRES	FEE
7	MULESHOE RANCH	AZ	49120	0
14	MCELWAIN-OLSEN	MA	66	0
13	TATKON	MA	40	0
9	DAVID H. SMITH	MA	830	0
11	MIACOMET MOORS	MA	4	0
12	MOUNT PLANTAIN	MA	730	0
1	COMERTOWN PRAIRIE	MT	1130	0
2	PINE BUTTE SWAMP	MT	15000	0
10	HOFT FARM	MA	90	0

17. As seguintes cláusulas WHERE são logicamente equivalentes? Resposta: Não, as cláusulas não são logicamente equivalentes.

```
SELECT * FROM PRESERVE
WHERE NOT (ACRES < 50 AND STATE = 'MA')
```

Table 21: Displaying records 1 - 10

PNO	PNAME	STATE	ACRES	FEE
5	HASSAYAMPA RIVER	AZ	660	3
3	DANCING PRAIRIE	MT	680	0
7	MULESHOE RANCH	AZ	49120	0
40	SOUTH FORK MADISON	MT	121	0
14	MCELWAIN-OLSEN	MA	66	0
9	DAVID H. SMITH	MA	830	0
12	MOUNT PLANTAIN	MA	730	0
1	COMERTOWN PRAIRIE	MT	1130	0
2	PINE BUTTE SWAMP	MT	15000	0
80	RAMSEY CANYON	AZ	380	3

```
SELECT * FROM PRESERVE
WHERE NOT ACRES < 50 AND NOT STATE = 'MA'
```

Table 22: 8 records

PNO	PNAME	STATE	ACRES	FEE
5	HASSAYAMPA RIVER	AZ	660	3
3	DANCING PRAIRIE	MT	680	0
7	MULESHOE RANCH	AZ	49120	0
40	SOUTH FORK MADISON	MT	121	0
1	COMERTOWN PRAIRIE	MT	1130	0
2	PINE BUTTE SWAMP	MT	15000	0
80	RAMSEY CANYON	AZ	380	3
6	PAPAGONIA-SONOITA CREEK	AZ	1200	3

```
SELECT * FROM PRESERVE
WHERE ACRES >= 50 OR STATE <> 'MA'
```

Table 23: Displaying records 1 - 10

PNO	PNAME	STATE	ACRES	FEE
5	HASSAYAMPA RIVER	AZ	660	3
3	DANCING PRAIRIE	MT	680	0
7	MULESHOE RANCH	AZ	49120	0
40	SOUTH FORK MADISON	MT	121	0
14	MCELWAIN-OLSEN	MA	66	0
9	DAVID H. SMITH	MA	830	0
12	MOUNT PLANTAIN	MA	730	0
1	COMERTOWN PRAIRIE	MT	1130	0
2	PINE BUTTE SWAMP	MT	15000	0
80	RAMSEY CANYON	AZ	380	3

18. As seguintes cláusulas WHERE são logicamente equivalentes? Resposta: Não, as cláusulas não são logicamente equivalentes.

```
SELECT * FROM PRESERVE
WHERE NOT (ACRES < 50 OR STATE = 'MA')
```

Table 24: 8 records

PNO	PNAME	STATE	ACRES	FEE
5	HASSAYAMPA RIVER	AZ	660	3
3	DANCING PRAIRIE	MT	680	0
7	MULESHOE RANCH	AZ	49120	0
40	SOUTH FORK MADISON	MT	121	0
1	COMERTOWN PRAIRIE	MT	1130	0
2	PINE BUTTE SWAMP	MT	15000	0
80	RAMSEY CANYON	AZ	380	3
6	PAPAGONIA-SONOITA CREEK	AZ	1200	3

```
SELECT * FROM PRESERVE
WHERE NOT ACRES < 50 OR NOT STATE = 'MA'
```

Table 25: Displaying records 1 - 10

PNO	PNAME	STATE	ACRES	FEE
5	HASSAYAMPA RIVER	AZ	660	3

PNO	PNAME	STATE	ACRES	FEE
3	DANCING PRAIRIE	MT	680	0
7	MULESHOE RANCH	AZ	49120	0
40	SOUTH FORK MADISON	MT	121	0
14	MCELWAIN-OLSEN	MA	66	0
9	DAVID H. SMITH	MA	830	0
12	MOUNT PLANTAIN	MA	730	0
1	COMERTOWN PRAIRIE	MT	1130	0
2	PINE BUTTE SWAMP	MT	15000	0
80	RAMSEY CANYON	AZ	380	3

```
SELECT * FROM PRESERVE
WHERE NOT ACRES < 50 AND NOT STATE = 'MA'
```

Table 26: 8 records

PNO	PNAME	STATE	ACRES	FEE
5	HASSAYAMPA RIVER	AZ	660	3
3	DANCING PRAIRIE	MT	680	0
7	MULESHOE RANCH	AZ	49120	0
40	SOUTH FORK MADISON	MT	121	0
1	COMERTOWN PRAIRIE	MT	1130	0
2	PINE BUTTE SWAMP	MT	15000	0
80	RAMSEY CANYON	AZ	380	3
6	PAPAGONIA-SONOITA CREEK	AZ	1200	3

```
Sys.setenv(TZ = "America/Sao_Paulo")
Sys.time()
```

```
[1] "2026-08-31 08:29:29 -03"
```